



Visvesvaraya Technological University (VTU)

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Module 5

TRENDS IN BIOENGINEERING (QUALITATIVE):

Module – 5

TRENDS IN BIOENGINEERING (QUALITATIVE):

Bioprinting techniques and materials, 3D printing of ear, bone and skin. 3D printed foods. Electrical tongue and electrical nose in food science, DNA origami and Biocomputing, Bioimaging and Artificial Intelligence for disease diagnosis. Self-healing Bioconcrete (based on bacillus spores, calcium lactate nutrients and biomineralization processes) and Bioremediation and Biomining via microbial surface adsorption (removal of heavy metals like Lead, Cadmium, Mercury, Arsenic).

DNA ORIGAMI AND BIOCOMPUTING:

Biological materials are self-assembled with near-atomic precision in living cells, whereas synthetic 3D structures generally lack such precision and controllability. Recently, DNA nanotechnology, especially DNA origami technology, has been useful in the bottom-up fabrication of well-defined nanostructures ranging from tens of nanometers to sub-micrometers. In this Primer, we summarize the methodologies of DNA origami technology, including origami design, synthesis, functionalization and characterization. We highlight applications of origami structures in nanofabrication, nano photonics and nanoelectronics, catalysis, computation, molecular machines, bioimaging, drug delivery and biophysics. We identify challenges for the field, including size limits, stability issues and the scale of production, and discuss their possible

solutions. We further provide an outlook on next-generation DNA origami techniques that will allow in vivo synthesis and multiscale manufacturing.

DNA origami is the nanoscale folding of DNA to create arbitrary two- and three-dimensional shapes at the nanoscale. The specificity of the interactions between complementary base pairs makes DNA a useful construction material, through design of its base sequences. DNA is a wellunderstood material that is suitable for creating scaffolds that hold other molecules in place or to create structures all on its own.

The current method of DNA origami was developed by Paul Rothemund at the California Institute of Technology. The process involves the folding of a long single strand of viral DNA (typically the 7,249 bp genomic DNA of M13 bacteriophage) aided by multiple smaller "staple" strands. These shorter strands bind the longer in various places, resulting in the formation of a pre-defined two- or three-dimensional shape. Examples include a smiley face and a coarse map of China and the Americas, along with many three-dimensional structures such as cubes.

To produce a desired shape, images are drawn with a raster fill of a single long DNA molecule. This design is then fed into a computer program that calculates the placement of individual staple strands. Each staple binds to a specific region of the DNA template, and thus due to Watson-Crick base pairing, the necessary sequences of all staple strands are known and displayed. The DNA is mixed, then heated and cooled. As the DNA cools, the various staples pull the long strand into

the desired shape. Designs are directly observable via several methods, including electron microscopy, atomic force microscopy, or fluorescence microscopy when DNA is coupled to fluorescent materials.

Bottom-up self-assembly methods are considered promising alternatives that offer cheap, parallel synthesis of nanostructures under relatively mild conditions.

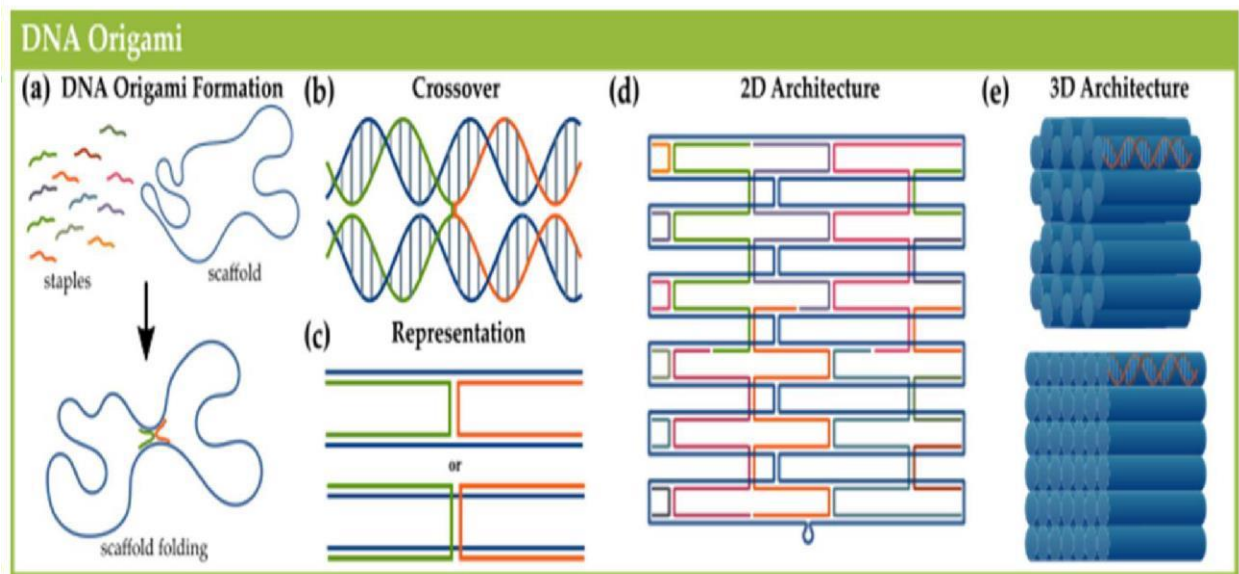
Since the creation of this method, software was developed to assist the process using CAD software. This allows researchers to use a computer to determine the way to create the correct staples needed to form a certain shape. One such software called caDNAno is an open source software for creating such structures from DNA. The use of software has not only increased the ease of the process but has also drastically reduced the errors made by manual calculations.

Applications:

Many potential applications have been suggested in literature, including enzyme immobilization, drug delivery systems, and nanotechnological self-assembly of materials. Though DNA is not the natural choice for building active structures for nanorobotic applications, due to its lack of structural and catalytic versatility, several papers have examined the possibility of molecular walkers on origami and switches for algorithmic computing. The following paragraphs list some of the reported applications conducted in the laboratories with clinical potential.

Long strands of DNA are folded into a complex scaffold of staple strands having 200–300 nucleotides. This leads to formation of a complex structure that has

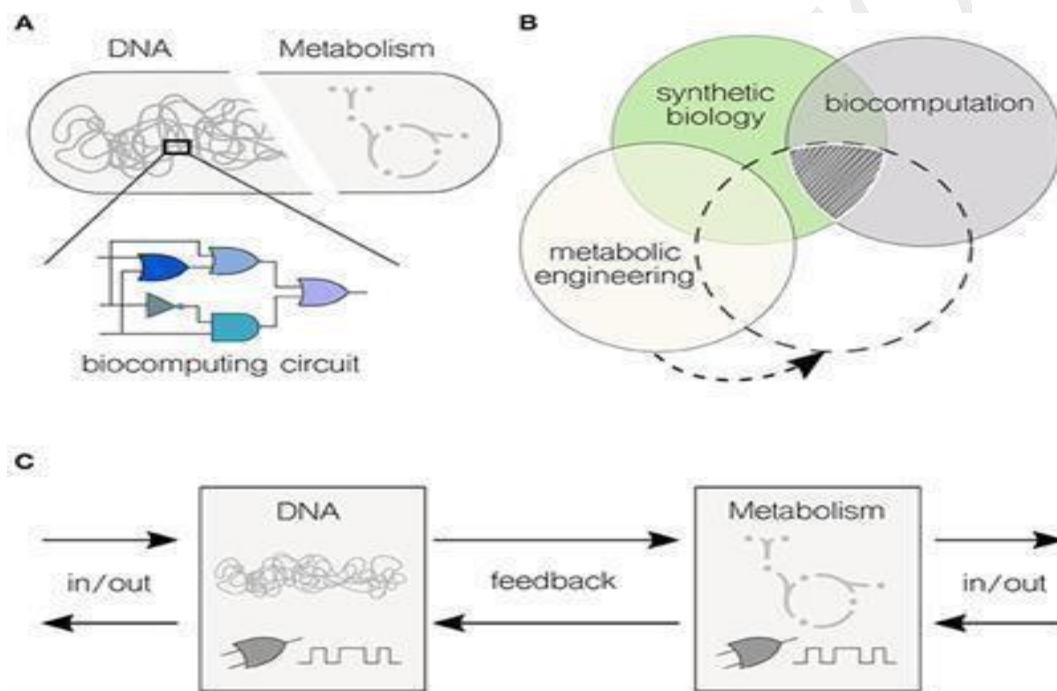
characteristic features because of their nanoscale dimensions. These DNA nanostructures are known to still be in their preliminary developmental stages, since key domains, such as their biocompatibility and physiochemical characterizations are yet to be established. However, theoretically, DNA origami has the immense potential to contribute significantly in a wide range of fields, such as diagnosis and drug delivery. Cancer therapy and diagnosis is one such potential domain where DNA origami showed significant anticancer efficacy and may contribute immensely.



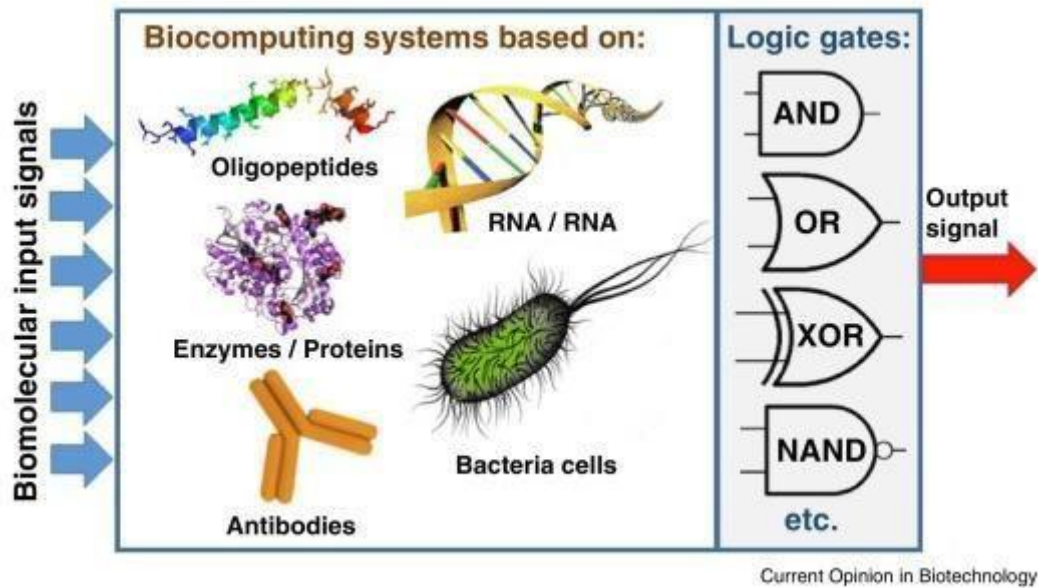
BIOCOMPUTING:

A computer that uses components of biological origin (such as molecules of DNA) instead of electrical components. The device is rudimentary—it can only perform basic high-school-level math problems.

In the quest to understand and model the healthy or sick human body, researchers and medical doctors are utilizing more and more quantitative tools and techniques. This trend is pushing the envelope of a new field we call Biomedical Computing, as an exciting frontier among signal processing, pattern recognition, optimization, nonlinear dynamics, computer science and biology, chemistry and medicine.



Computing process which use synthesized biological components to store and manipulate data analogous to processes in the human body. The result is small, faster computing processes that operates with great accuracy. Main component used is DNA. The main application is in disease prediction and disease diagnosis.



BIOIMAGING AND ARTIFICIAL INTELLIGENCE FOR DISEASE

DIAGNOSIS:

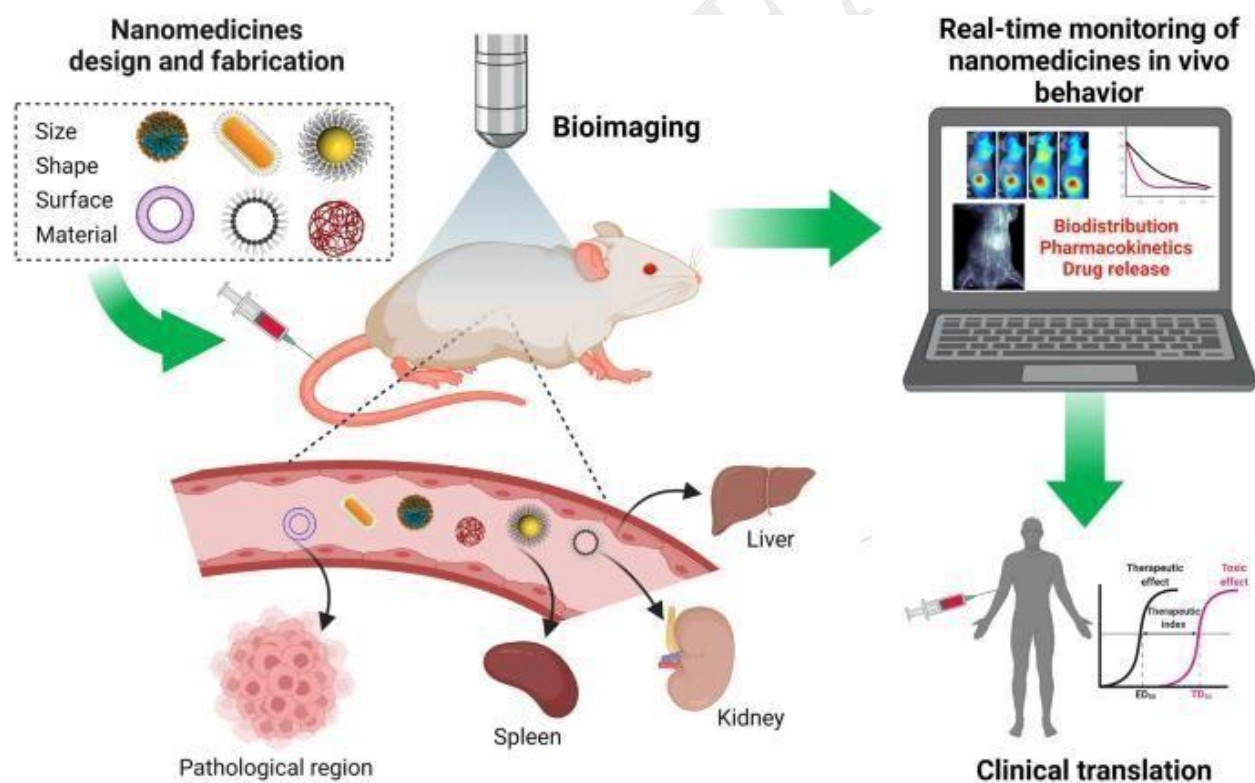
1) BIOIMAGING:

Bioimaging is a noninvasive process of visualizing biological activity in a specific period. It does not inhibit the various life processes such as movement, respiration, etc., and it helps to report the 3D structure of specimens apart from inferencing physically. It is helpful in connecting the observation of subcellular structures and all the tissues in the multicellular organisms.

The imaging of biological samples, or bioimaging, plays a key role in current life science research, enabling scientists to analyze molecules, cells and tissues from a range of living systems.

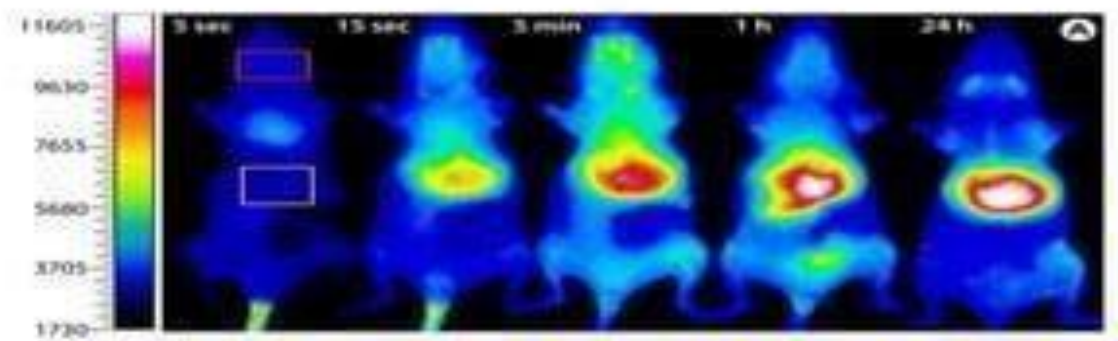
Nanoparticle fluorescence imaging has been used in gene detection, protein analysis, enzyme activity evaluation, element tracing, cell tracking, early stage disease diagnosis, tumor related research, and monitoring real time therapeutic effects.

Bioimaging spans the observation of subcellular structures and entire cells over tissues up to entire multicellular organisms. Among others, it uses light, fluorescence, electrons, ultrasound, X-ray, magnetic resonance and positrons as sources for imaging.



Importance of Bioimaging:

- It allows in vivo imaging of biological processes, including cellular signaling and interactions and the movement of molecules through membranes.
- bioimaging offers precise tracking of metabolites that can be used as biomarkers for disease identification, progress and treatment response.



2) ARTIFICIAL INTELLIGENCE IN DISEASE DIAGNOSIS:

Artificial intelligence can assist providers in a variety of patient care and intelligent health systems. Artificial intelligence techniques ranging from machine learning to deep learning are prevalent in healthcare for disease diagnosis, drug discovery, and patient risk identification. Numerous medical data sources are required to perfectly diagnose diseases using artificial intelligence techniques, such as ultrasound, magnetic resonance imaging, mammography, genomics, computed tomography scan, etc. Furthermore, artificial intelligence primarily enhanced the infirmity experience and sped up preparing patients to continue their rehabilitation at home.

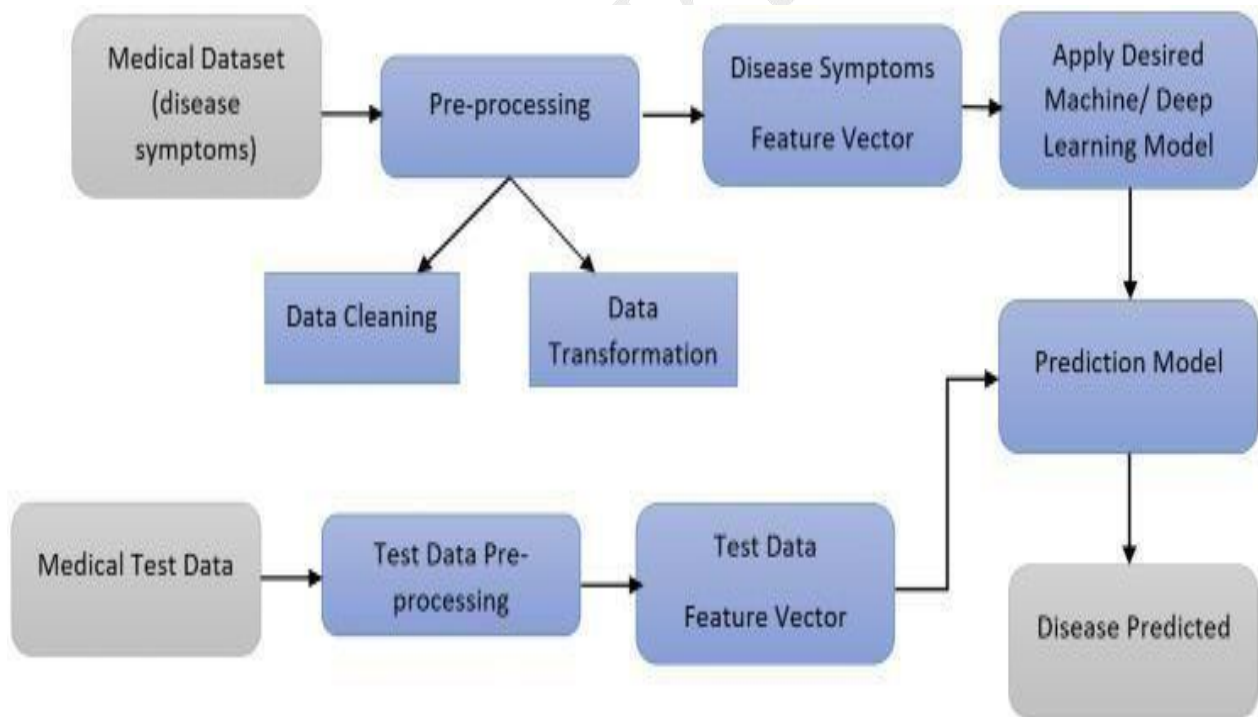
Detecting any irresistible ailment is nearly an afterward movement and forestalling its spread requires ongoing data and examination. Hence, acting rapidly with accurate data tosses a significant effect on the lives of individuals around the globe socially and financially (Minaee et al. 2020). The best thing about applying AI in health care is to improve from gathering and processing valuable data to programming surgeon robots.

AI describes the capability of a machine to study the way a human learns, e.g., through image identification and detecting pattern in a problematic situation. AI in health care alters how information gets composed, analysed, and developed for patient care.

System planning is the fundamental abstract design of the system. It includes the framework's views, the course of action of the framework, and how the framework carries on underneath clear conditions. A solid grip of the framework design can help the client realize the limits and boundaries of the said framework. In pre-preparing, real-world information requires upkeep and pre-preparing before being taken care of by the calculation. Because of the justifiable explanation, real-world data regularly contains mistakes regarding the utilized measures yet cannot practice such blunders. Accordingly, information pre-preparing takes this crude information, cycles it, eliminates errors, and spares it an extra examination. Information experiences a progression of steps during pre-handling: Information is purged by various strategies in information cleaning. These strategies involve gathering information, such as filling the information spaces that are left clear or decreasing information, such as the disposal of

commas or other obscure characters. In information osmosis, the information is joined from a combination of sources.

The information is then amended for any blend of mistakes, and they are quickly taken care of. Information Alteration: Data in this progression is standardized, which depends upon the given calculation. Information standardization can be executed utilizing several ways. This progression is obligatory in most information mining calculations, as the information wants to be as perfect as possible. Information is then mutual and developed.



SELF HEALING BIOCONCRETE:

Bio-concrete is a self-healing form of concrete designed to repair its own cracks.

To heal cracks in the concrete, Jonkers chose bacteria (*Bacillus pseudo-rmus* and *B. cohnii*), that are able to produce limestone on a biological basis. The positive side-effect of this property: the bacteria consume oxygen, which in turn prevents the internal corrosion of reinforced concrete. However, the bacteria do not pose a risk to human health, since they can only survive under the alkaline conditions inside the concrete. Based on these findings, Jonkers and his team of researchers developed three different bacterial concrete mixtures: self-healing concrete, repair mortar, and a liquid repair system.

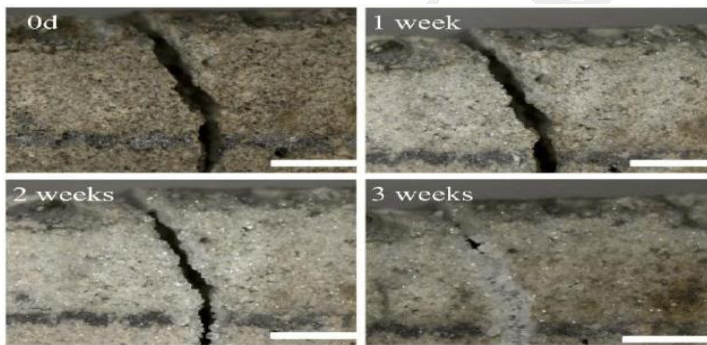
In self-healing concrete, bacterial content is integrated during construction, while the repair mortar and liquid system only come into play when acute damage has occurred on concrete elements. Self-healing concrete is the most complex of the three variants. Bacterial spores are encapsulated within two-to four-millimeter wide clay pellets and added to the cement mix with separate nitrogen, phosphorous and a nutrient agent. This innovative approach ensures that bacteria can remain dormant in the concrete for up to 200 years. Contact with nutrients occurs only if water penetrates into a crack – and not while mixing cement. This variant is well-suited for structures that are exposed to weathering, as well as points that are difficult to access for repair workers. Thus, the need for expensive and complex manual repairs is eliminated.

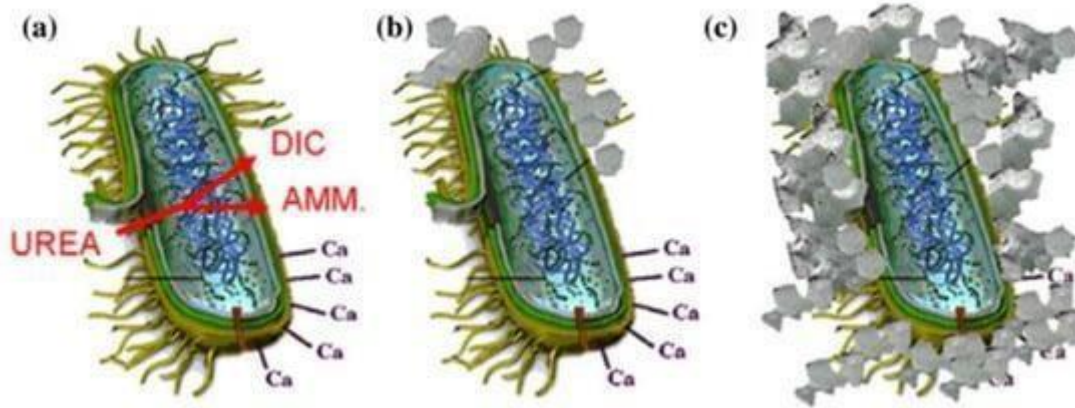
Self-healing concrete is nothing but concrete which can retain itself to the original state when it is subjected to cracks." Bio-concrete is a material that will biologically produce minerals like limestone with the help of bacteria present in it, which will heal cracks that appear on the concrete surfaces. Bacterial self-

healing is an innovative technology allowing repairing open micro-cracks in concrete by CaCO_3 precipitation. This bio-technology improves the durability of the structure. In this paper, peptone, yeast extract and *Bacillus Subtilis* were added as microbial adjuvant in concrete mix design.

Rahbar predicts self-healing concrete could extend the life of a structure from 20 years, for example, to 80 years. Other research into creating self-healing concrete has focused on adding microbes and *Bacillus megaterium*, a spore-forming bacteria that produces an enzyme that is expelled into the concrete mix.

The healing agent consisting of *B. cohnii* spores, calcium lactate and yeast extract immobilized in light-weight aggregates was also combined with cement, fly ash, limestone powder, PVA fibers, water in a repair mortar.





BIOREMEDIATION AND BIO MINING VIA MICROBIAL SURFACE

ADSORPTION:

1) BIOREMEDIATION:

Bioremediation is a biotechnical process, which abates or cleans up contamination. It is a type of waste management technique which involves the use of organisms to remove or utilize the pollutants from a polluted area. Types of Bioremediation .

Bioremediation is of three types –

1) Biostimulation:

As the name suggests, the bacteria is stimulated to initiate the process. The contaminated soil is first mixed with special nutrients substances including other vital components either in the form of liquid or gas. It stimulates the growth of microbes thus resulting in efficient and quick removal of contaminants by microbes and other bacterias.

2) Bioaugmentation:

At times, there are certain sites where microorganisms are required to extract the contaminants. For example – municipal wastewater. In these special cases, the process of bioaugmentation is used. There's only one major drawback in this process. It almost becomes impossible to control the growth of microorganisms in the process of removing the contaminant.

3) Intrinsic Bioremediation:

The process of intrinsic bioremediation is most effective in the soil and water because of these two biomes which always have a high probability of being full of contaminants and toxins. The process of intrinsic bioremediation is mostly used in underground places like underground petroleum tanks. In such place, it is difficult to detect a leakage and contaminants and toxins can find their way to enter through these leaks and contaminate the petrol. Thus, only microorganisms can remove the toxins and clean the tanks.

Bioremediation helps clean up water sources, create healthier soil, and improve air quality around the globe. But unlike excavation-based remediation processes, which can be disruptive, bioremediation is less intrusive and can facilitate remediation of environmental impacts without damaging delicate ecosystems.

Immobilization of microbial cells and enzymes by adsorption takes place through their physical interaction with the surface of water-insoluble carriers. This method, commonly used in bioremediation processes, is quick, simple, eco-friendly and cost-effective.

Microorganisms are utilized in bioremediation because of their ability to degrade environmental pollutants due to their metabolism via biochemical pathways related to the organism's activity and growth.

2) BIOMINING:

Biomining is the process of using microorganisms (microbes) to extract metals of economic interest from rock ores or mine waste. Biomining techniques may also be used to clean up sites that have been polluted with metals.

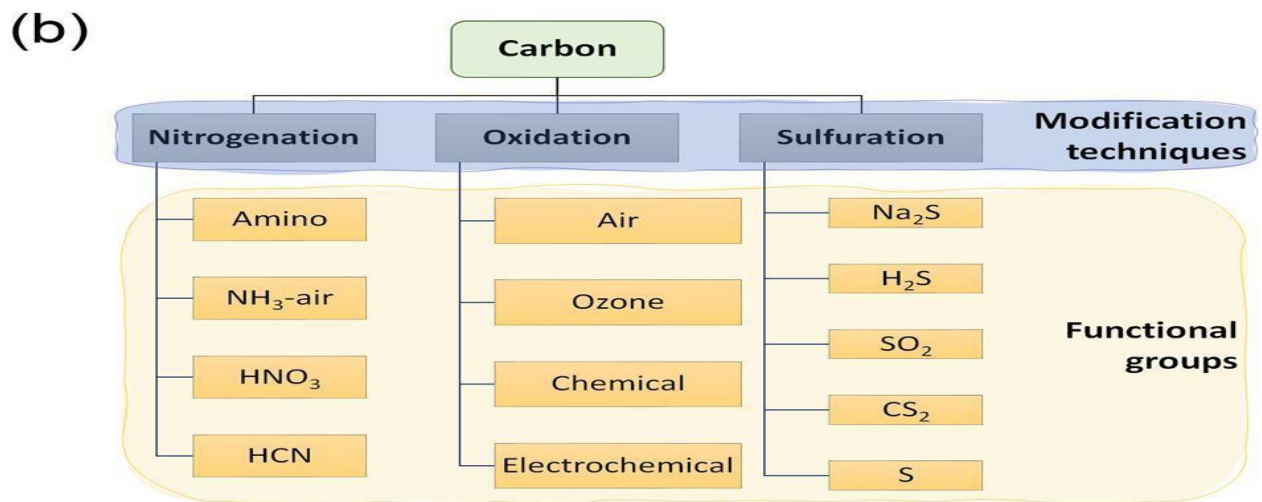
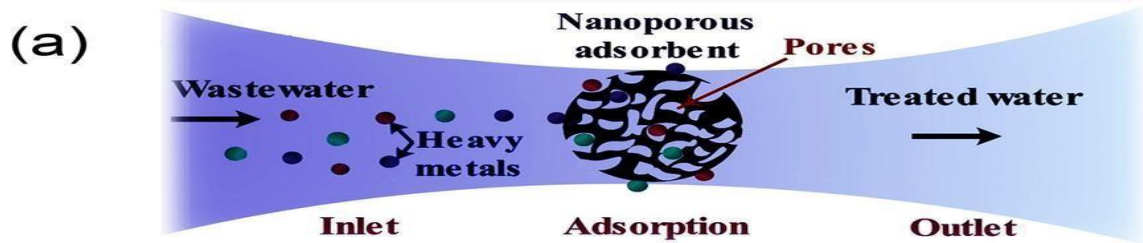
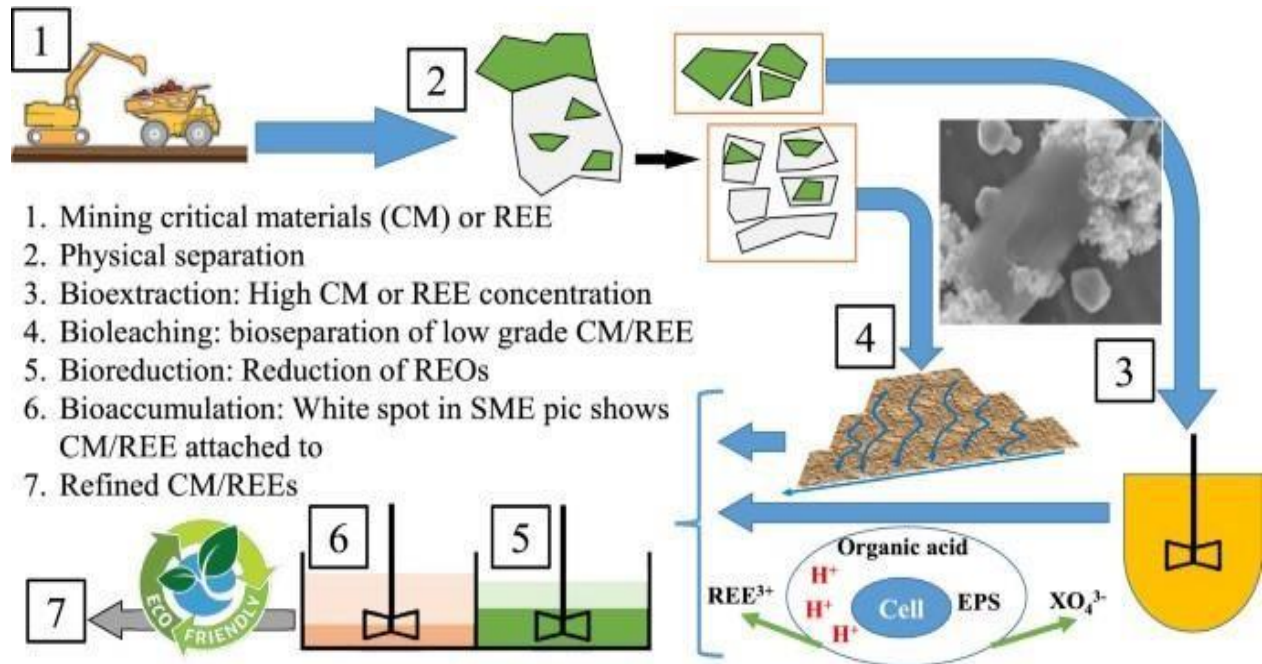
Valuable metals are commonly bound up in solid minerals. Some microbes can oxidize those metals, allowing them to dissolve in water. This is the basic process behind most biomining, which is used for metals that can be more easily recovered when dissolved than from the solid rocks. A different biomining technique, for metals which are not dissolved by the microbes, uses microbes to

break down the surrounding minerals, making it easier to recover the metal of interest directly from the remaining rock.

Most current biomining operations target valuable metals like copper, uranium, nickel, and gold that are commonly found in sulfidic (sulfur-bearing) minerals. Microbes are especially good at oxidizing sulfidic minerals, converting metals like iron and copper into forms that can dissolve more easily. Other metals, like gold, are not directly dissolved by this microbial process, but are made more accessible to traditional mining techniques because the minerals surrounding these metals are dissolved and removed by microbial processes. When the metal of interest is directly dissolved, the biomining process is called “bioleaching,” and when the metal of interest is made more accessible or “enriched” in the material left behind, it is called “biooxidation.” Both processes involve microbial reactions that can happen anywhere the microbes, rocks, and necessary nutrients, like oxygen, occur together.

Bioleaching (or biomining) is a process in mining and biohydrometallurgy (natural processes of interactions between microbes and minerals) that extracts valuable metals from a low-grade ore with the help of microorganisms such as bacteria or archaea.

Instead of separating the metal from the pyrite with high temperatures or pressures, biomining uses microbes from the *Acidithiobacillus* and *Leptospirillum* genera to do the job.



- A) Heavy metal ions adsorption process; the metal ions of wastewater adhere to the surface of nanoporous adsorbents, which has a high surface area due to its porosity. The adsorption process could be selective for one or more metals than others. The regeneration process could be achieved using a desorbing agent.
- B) Various modification techniques (i.e., nitrogenation, oxidation, and sulfuration) are used to functionalize carbon with different functional groups. Functionalization enhances adsorption capacity and stability.

Module-05

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